

Big Data Analytics

Big data analytics is the process of examining large and varied data sets --i.e., big data -- to uncover **hidden patterns, unknown correlations, market trends, customer preferences** and other useful information that can help organizations make more-informed business decisions.

Analytical finding can lead to more effective **marketing, better customer service & satisfaction, newer products & services, improved operational efficiency, reduced expenditure** etc.. (ex: Amazon recommendations)

Recommendation systems in ecommerce

- Make your marketing unique to every customer
- Learn about your audience and tailor your messaging to create personalized experiences for all your customers.
- Suppose a customer buys new running shoes every three months. They always buy the same brand. If this customer comes back to the site and searches for running shoes, it's a good guess that they will buy their favorite brand again—but maybe they would like new features, a different color, or a shoe with new product endorsement.
- A personalized product recommendation for this customer keeps old choices available, shows the possibility of a "new and improved" customer experience at a different price point first.

Recommendation systems in ecommerce

• Recommendation engines usually look at data such as:

- A customer's search queries
- Their purchase history
- What's currently in their shopping cart
- Their social behavior (likes, shares, etc.)
- Their geographic location
- The customer's audience segment (demographics)

Data → Raw facts collected from multiple sources (databases, logs, social media, etc.).

Information → Processed and structured form of data, presented in reports, dashboards, or charts.

Insight → Analysis of information to identify patterns, trends, comparisons, and conclusions.

Actionable Insight →

The most valuable form of insight.

Provides clear, practical direction.

Encourages immediate decisions and actions.

Delivers maximum business value from data analytics.

.Actionable insights are insights that deliver immense value, provide a clear direction, and drive instant actions towards achieving business objectives.

. These insights are the most valuable and provide **maximum returns on data analytics investments.**

.Actionable insights are distilled from raw data in the following way.

.Data: Raw data is collected from multiple sources such as information databases, spreadsheets, plans, social media comments, call logs, and so on. This data can be structured (forms and tables) or unstructured (log files, comments) and qualitative (observations) or quantitative (numbers or units).

. Information: Raw data, when processed, categorized, and organized, is converted into information. This information can be presented in the form of reports, dashboards, charts, and other visualizations.

.Insight: Information is analyzed to identify patterns, summarize findings, and draw conclusions in the form of insights. Insights help to understand business events, answer questions, compare details, spot similarities and differences, group similar items, and track trends.

.Actionable Insight: Among the various insights that emerge from data, actionable insights are crucial. These insights encourage informed actions and decisions based on data-driven recommendations. The control to take action or not rests with the user.

Benefits of Actionable Insights

.Enhance Decision Making: Actionable insights give business users the much-needed support to enhance their decision-making. When insights are aligned with an organization's business objectives and KPIs, they become more actionable and influence better decision-making.

.Actionable insights provide decision intelligence on what matters most to decision-makers based on their role, work, or goal.

Benefits of Actionable Insights

.Understand Context Better: Actionable insights go beyond answers to provide context for changing business metrics.

.For example, the monthly report shows a surge in sales by 20% over the previous month. It may look good now, but it is missing a context.

. The surge was due to an increase in the price of goods, while the number of transactions actually went down.

. Such a surge might not help in the long run. This actionable insight alerts the management that the price rise is driving away customers, and they can take measures to win back customers.

. Actionable insights explain not only what happened but also why it happened and how to proceed ahead.

Gain Clarity

- While data processing and insight extraction can be handled using automated analytics, machine learning, and AI algorithms, the true value comes from **presenting those insights clearly**.
- Actionable insights become more meaningful and useful when displayed in **interactive charts and visualizations**, allowing users to explore the data in ways that best suit their needs.

Get Specific Recommendations

Actionable insights filter out the **noise from lengthy reports, multiple dashboards**, and overwhelming data to provide **clear and focused guidance**.

For example,

- An actionable insight highlighting **growing demand from a new customer segment**,
- Presented as an automated business headline,
- Enables customer success managers to **develop targeted offerings and strengthen customer relationships**.

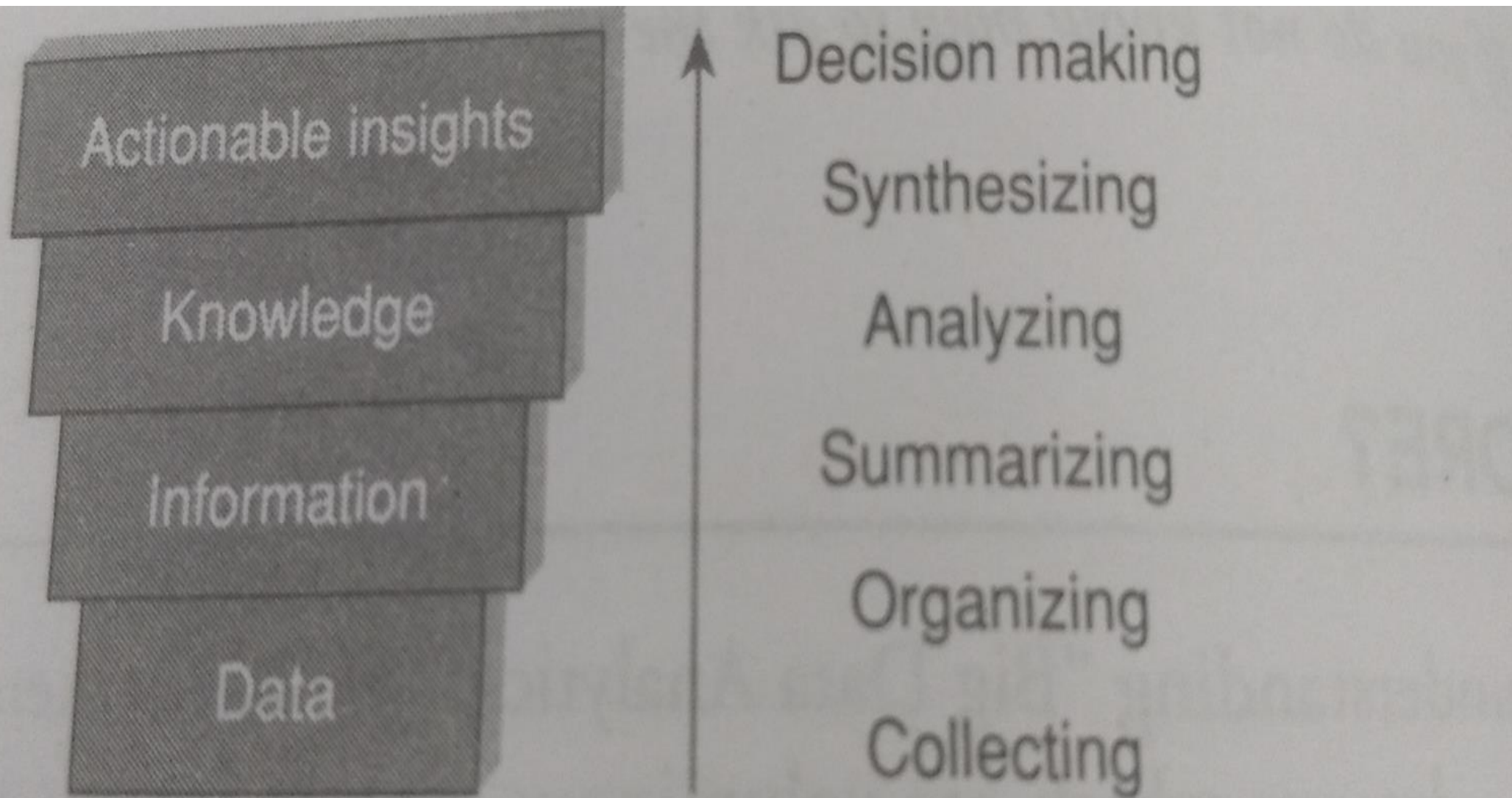
Save Significant Time and Efforts:

.When users get access to relevant, focused, and personalized insights, they can understand business situations accurately and make quick decisions.

. With actionable insights, users no longer have to go through volumes of information and unnecessary vanity insights.

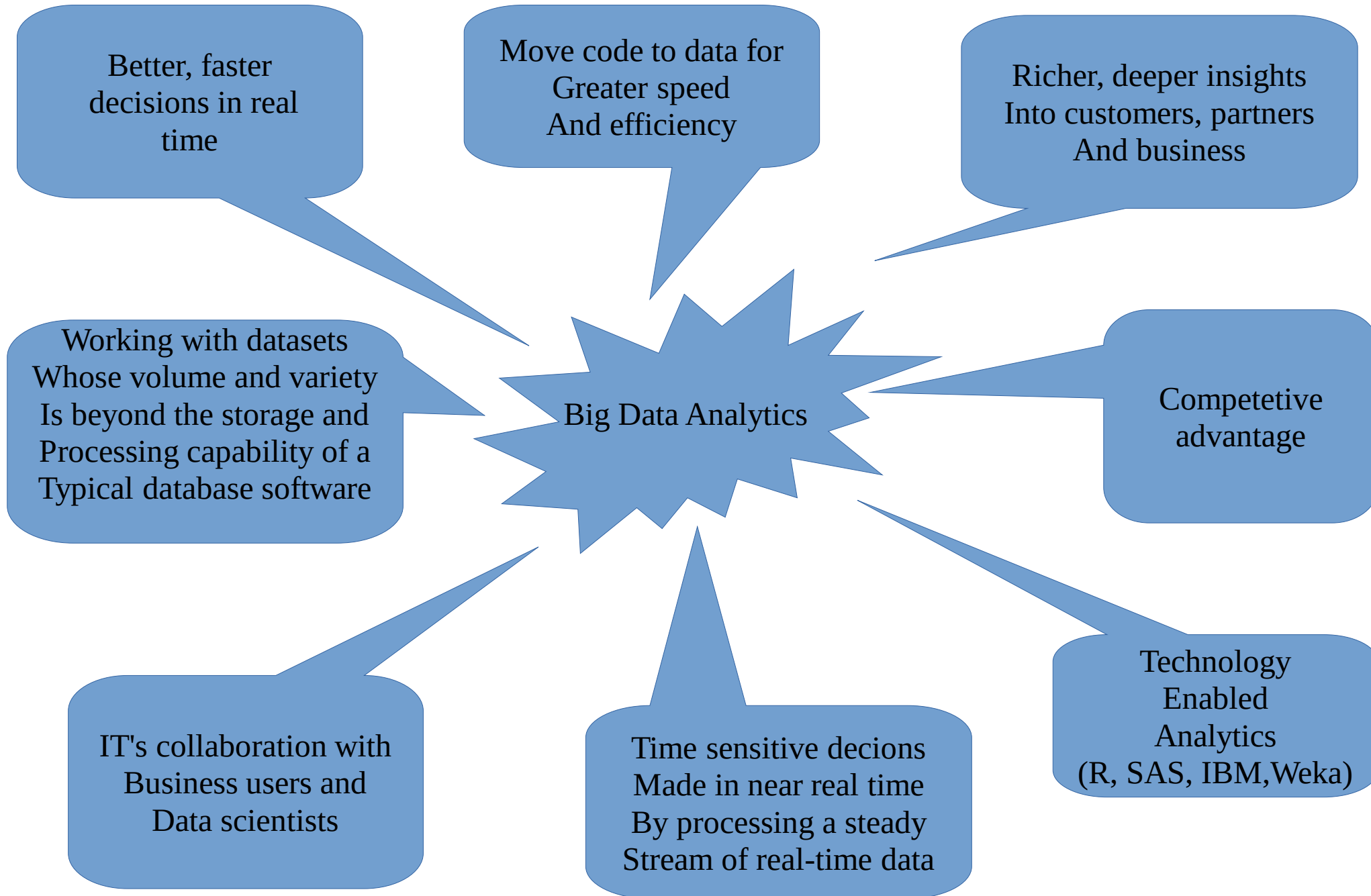
. With modern data analytics platforms users can access data easily and get insights in an automated way without investing time and effort in learning complex techniques or depending on experts.

Where Do We Begin ?

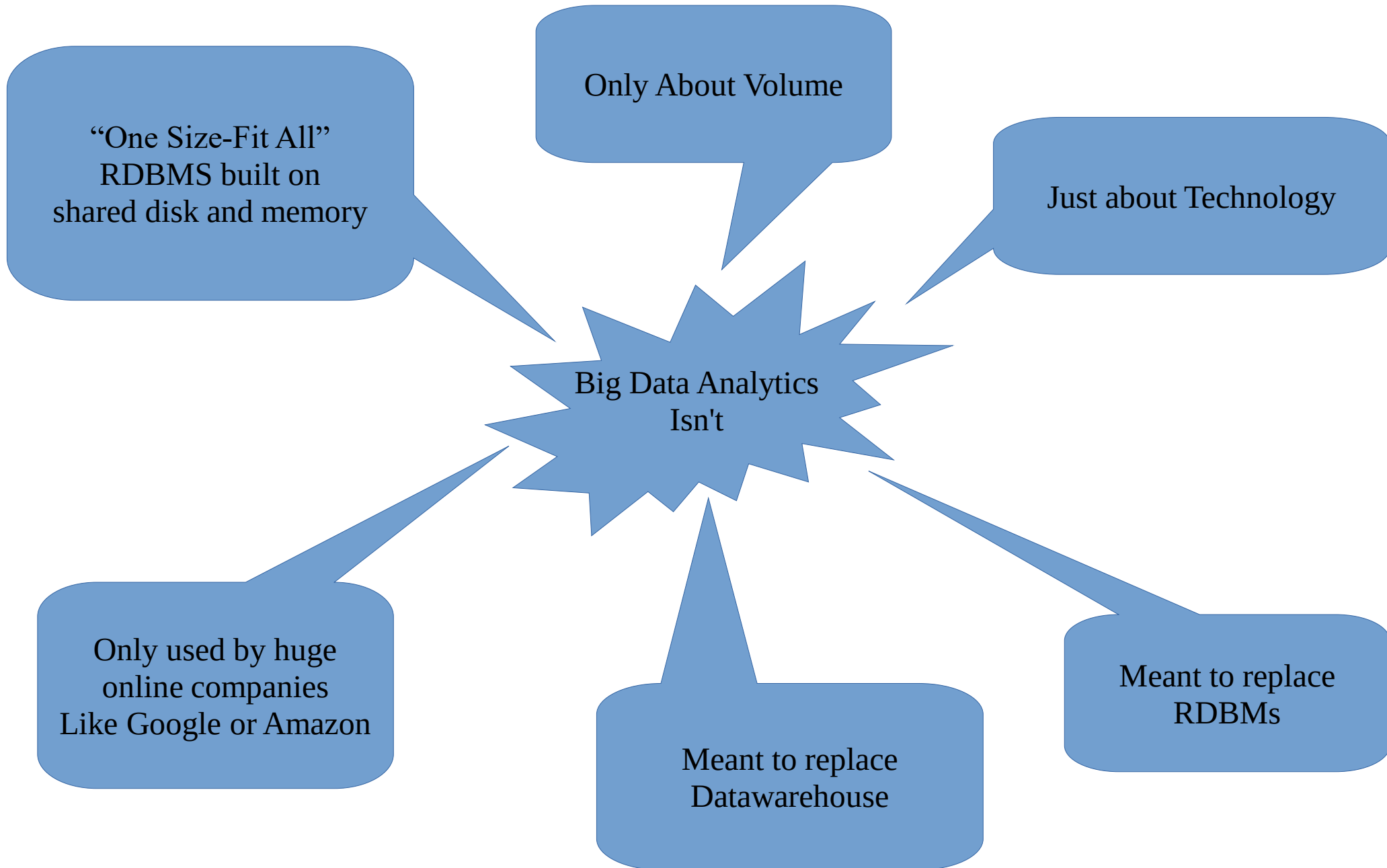


Transformation of data to yield actionable insights

What is Big Data Analytics?



What is Big Data Analytics isn't?



Why sudden hype around BDA ?

- Data growing at 40% compound rate, reaching nearly 50 ZB by 2020
- Cost per GB of storage has hugely dropped
- There is a shortage of user friendly analytical tools available in the market

Importance of Big Data Analytics

Better Decision-Making

- Organizations can analyze large volumes of structured, semi-structured, and unstructured data in real-time.
- It helps leaders make quicker and more accurate decisions based on facts rather than intuition.

•Improved Customer Experience

- By analyzing customer preferences, browsing history, and feedback, businesses can provide **personalized services**.
- Example: E-commerce platforms like Amazon recommend products using collaborative filtering.

•Risk Management & Fraud Detection

- Detects unusual patterns in transactions to prevent fraud (banking, insurance, healthcare).
- Identifies potential risks early and suggests preventive measures.

•Innovation & Product Development

- Insights from big data reveal **market trends and gaps**, helping companies design innovative products and services.
- Google and Netflix rely heavily on user data for new features.

•Operational Efficiency

- Helps streamline supply chain, production planning, and resource allocation.

Classification of Analytics

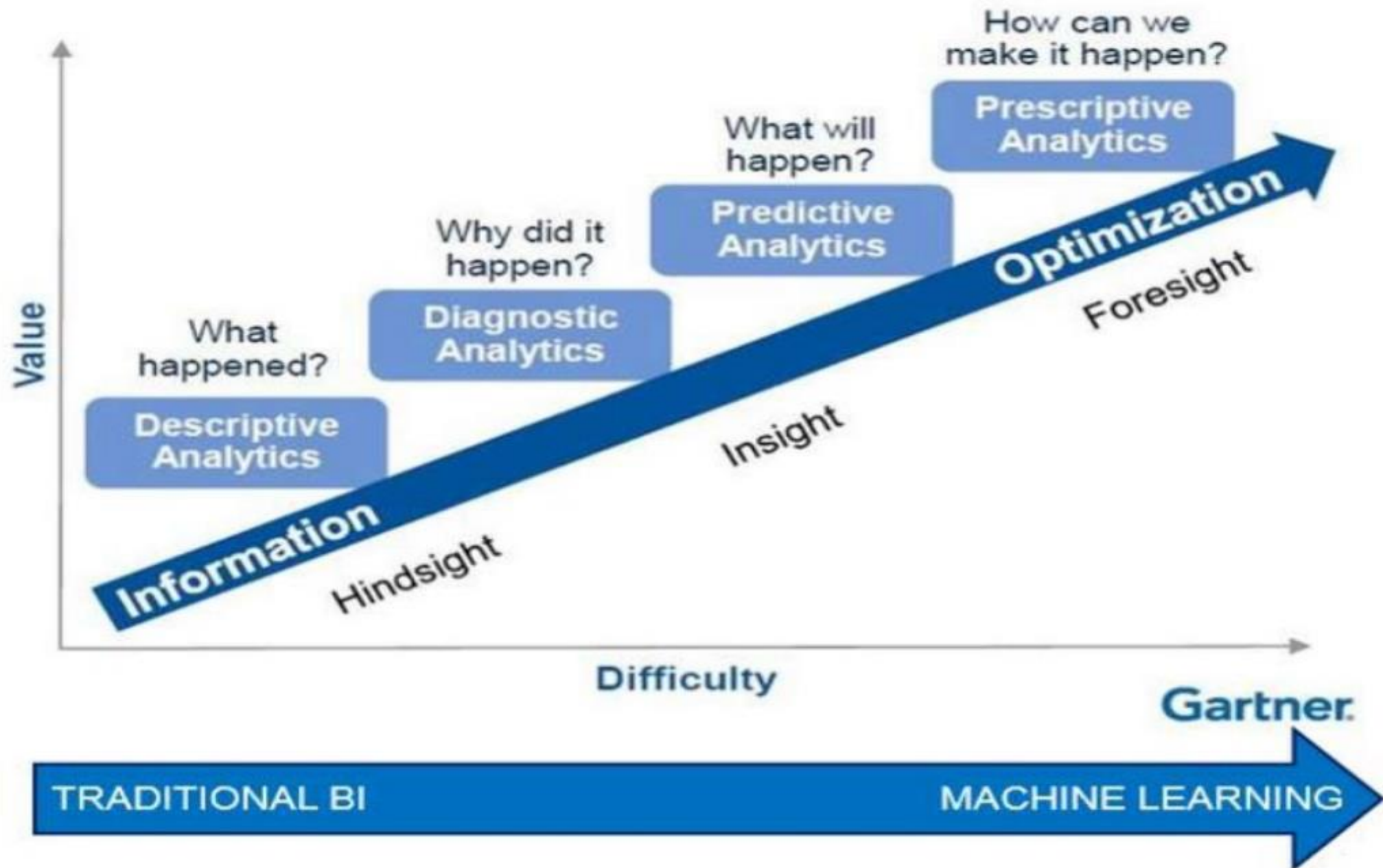
Analytics 1.0	Analytics 2.0	Analytics 3.0
Era: mid 1950's to 2009	Era: 2005 to 2012	Era: 2012 To Present
Descriptive Staistics (report on events, occurences etc of the past)	Descriptive+Predictive Statistics (use data from the past to made predictions for the future)	Descriptive+Predictive + Prescriptive Statistics (use data from the past to made predictions for the future at the same time make recommendations to leverage the situation to one's advantage)
Key Questions asked: What happed? Why did it happen ?	Key Questions asked: What will happen? Why will it happen?	Key Questions asked: What will happen? Why will it happen? When will it happen What should be the action taken to take advantage of what will happen?
Data from legacy systems, ERP,CRM and 3 rd party applications	Big Data	A blend of big data and data from legacy systems, ERP,CRM and 3 rd party applications

ERP* enterprise resource planning

CRM* customer relationship management.

Analytics 1.0	Analytics 2.0	Analytics 3.0
Small and structured data sources. Data stored in enterprise data warehouse or data marts	Big data is being taken up seriously. Data is mainly unstructured, arriving at a much higher pace.	A blend of big data and traditional analytics to yield insights and offerings with speed and impact
Data was internally sourced	Data was often externally sourced	Data is both being internally and externally sourced
Relational Databases	Database appliances, Hadoop clusters, SQL to Hadoop environments etc	In memory analytics, in database processing, agile analytical methods, machine learning techniques etc

Data Analytics - Classification



Type of Analytics	Question Answered	General Business Example	Healthcare Example
Descriptive Analytics	What Happened?	How many cars did we sell last year?	How many patients were diagnosed with HBP last year?
Diagnostic Analytics	Why Did It Happen?	Why did we only sell x cars last year?	Why did these patients develop HBP?
Predictive Analytics	What Will Happen?	If I run x advertising programs, how many cars can we sell?	What are the chances Mr. Jones' HBP will result in a stroke?
Prescriptive Analytics	How Can We Make it Happen?	What do we need to do to sell x number of cars?	Mr. Jones should be put on x medication to prevent his HBP from resulting in a stroke.

	Customer Analytics	Operational Analytics	People Analytics	Accounting Analytics
Descriptive Analytics	Who are my customers?	How is my operations running?	How are my peoples' performance?	How is my business performing?
Predictive Analytics	Which of my customers will churn?	How my operations will be impacted if some event happens?	Which people shall I promote? Which people will perform better?	If I invest in one area, how will the performance of the business be impacted?
Prescriptive Analytics	What offers shall I give to my potential churning?	What processes shall I change to minimize impact of disruption?	What incentives shall I give to improve performance?	What financial re-engineering I have to do improve performance

Greatest challenges that prevent business from capitalizing on big data

- Obtaining executive sponsorships for investments in big data and its related activities(such as training etc)
- Getting the business units to share information across organizational tower.
- Finding the right skills (business analysts and data scientists)
- Determining the approach to scale rapidly and elastically
- Deciding whether to use structured or unstructured, internal or external data to make business decisions
- Choosing the optimal way to report findings and analysis of big data for the presentations to make more sense
- Determining what to do with the insights created from big data

Top challenges facing Big data

- **Scale:** Should you scale vertically or horizontally ?
- **Security:** Most of big data platforms have poor security mechanisms
- **Schema:** Rigid schemas have no place. The need of the hour is dynamic schema
- **Continuous Availability:** How to provide 24/7 support
- **Consistency:** Should one opt for consistency or eventual consistency?
- **Partition tolerant:** How to build partition tolerant systems that can take care of both hardware and software failures?
- **Data quality:** how to maintain accuracy, completeness, timeliness etc

Terminologies used in Big Data Environments

1. In-Memory Analytics:

- In-memory analytics is a business intelligence (BI) methodology used to solve complex and time-sensitive business scenarios.
- It works by increasing the speed, performance and reliability when querying data.
- Business Intelligence deployments are typically disk-based, meaning the application queries data stored on physical disks. In contrast, with in-memory analytics, the queries and data reside in the server's random access memory (RAM)
- In-memory analytics helps improve the overall speed of a BI system and provides business-intelligence users with faster answers compared to traditional disk-based business intelligence, especially for queries that take a long time to process in a large database.

2. In-database processing:

- In-database processing, sometimes referred to as in-database analytics, refers to the integration of data analytics into data warehousing functionality.
- In-database analytics is a technology that allows data processing to be conducted within the database by building analytic logic into the database itself.
- Doing so eliminates the time and effort required to transform data and move it back and forth between a database and a separate analytics application.

Data doesn't have to be extracted, transformed, and reloaded (ETL) for analysis.

Everything happens where the data already resides → faster and more efficient

This technology is widely adopted in large databases, including those applied in credit card fraud detection

3. Symmetric Multiprocessor System (SMP):

Symmetric multiprocessing (SMP) is a **multiprocessor computer architecture** where:

- Two or more **identical processors** are connected to a **single shared memory**.
- All processors have equal access to **I/O devices** and system resources.
- A **single operating system** controls and treats all processors equally (none has higher priority).
- This is the most common architecture used in **modern multiprocessor systems**, including **multi-core CPUs** (where each core is treated like a separate processor).

• **Processors (P1, P2, ... Pn)** → Multiple identical CPUs that execute tasks.

• **Cache** → Each processor has its own cache to store frequently used data.

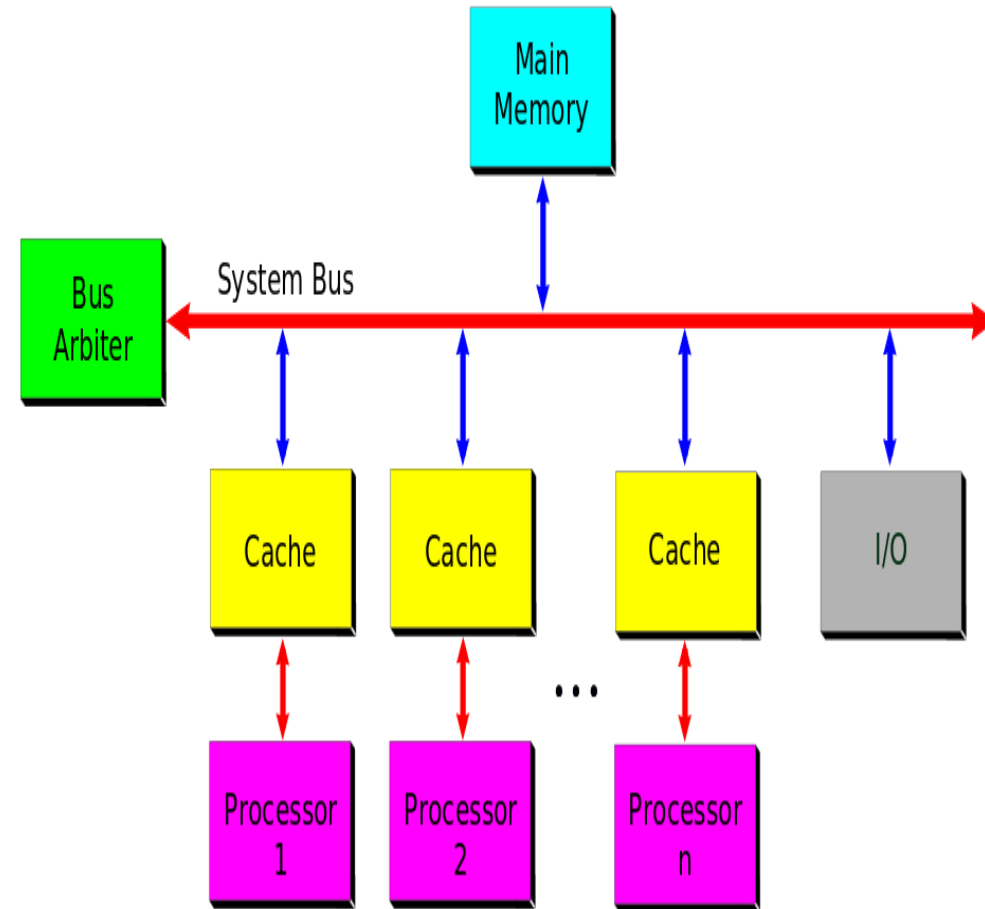
• **System Bus** → A common pathway that connects all processors, memory, and I/O.

• **Main Memory** → Shared memory accessible to all processors.

• **I/O devices** → Shared input/output resources accessible by all CPUs.

• **Bus Arbiter** → Manages access to the system bus, ensuring no two processors use it simultaneously.

SMP - Symmetric Multiprocessor System



- Each processor executes its own program independently.
- They can all access **shared memory** and **I/O devices** via the system bus.
- The **bus arbiter** prevents conflicts when multiple processors attempt to access the bus at the same time.
- Since processors are identical, the **operating system can schedule any process on any CPU**, improving performance.

Characteristics:

Tightly coupled system – All processors are connected and work closely together.

Homogeneous processors – All CPUs are identical and capable of performing any task.

Shared resources – Memory, I/O devices, and interrupt systems are common to all processors.

System Bus / Crossbar – Communication between processors and memory/I/O is done through a shared bus or switching mechanism.

4. Massively Parallel Processing (MPP):

- **MPP (Massively Parallel Processing)** means breaking a big program into smaller parts and running them **in parallel** on **many processors**.
 - Each processor has its **own memory, cache, I/O system, and even its own operating system**.
 - The processors (nodes) don't share memory directly — they communicate using an **interconnection network** and a **messaging interface**.
 - This is why MPP is called a “**shared-nothing architecture**” (unlike SMP where memory and I/O are shared).
-
- **Independent Processors** → Each has its own CPU + Cache.
 - **Private Memory & I/O** → No processor shares memory or storage with another.
 - **Interconnection Network** → A communication link between processors.
 - **Independent OS** → Each node can run its own operating system instance.
 - **Loosely Coupled** → Each processor works independently, only communicating when necessary.

Data Warehouses

Examples: **Amazon Redshift, Teradata, Google BigQuery**.

When you run a query on billions of records:

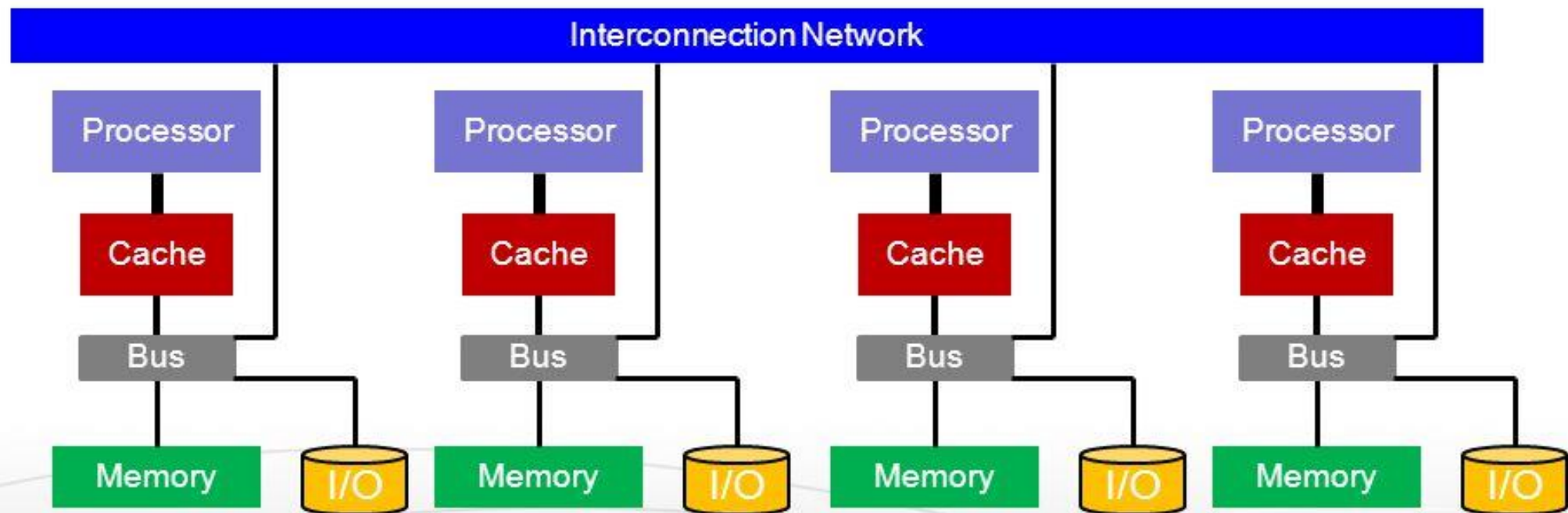
Database is split across multiple processors.

Each processor works on a portion of the data stored in its memory.

Results are combined and returned very quickly.

Massively Parallel Processors

- Massively Parallel Processors (MPP) architecture consists of nodes with each having its own processor, memory and I/O subsystem



- An independent OS runs at each node

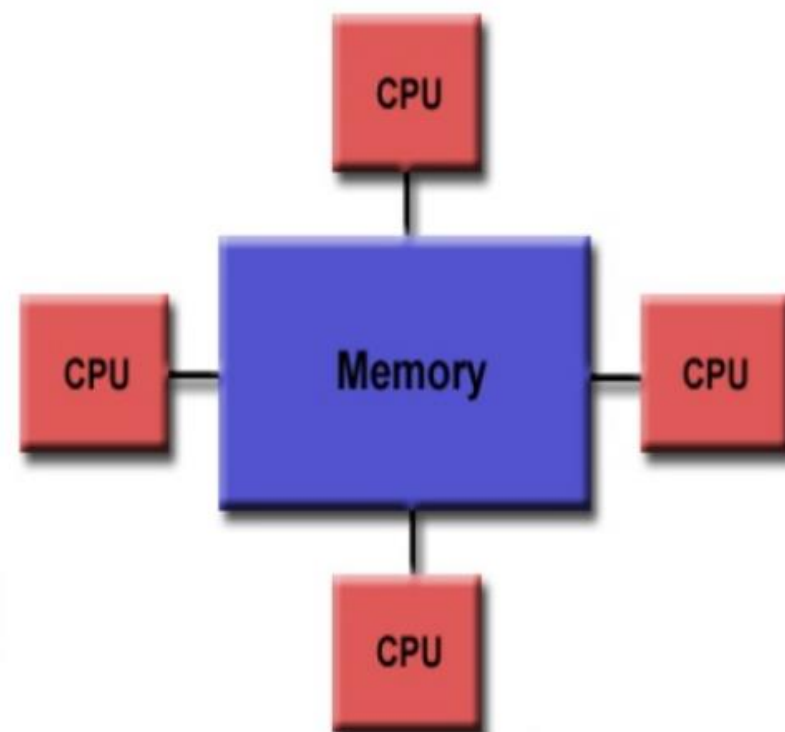
5. Difference between Parellel and Distributed systems

Parallel system:

- A system is said to be a **Parallel System** in which multiple processor have direct access to shared memory which forms a common address space.
- Usually tightly-coupled system are referred to as **Parallel System**. In these systems, there is a single system wide primary memory (address space) that is shared by all the processors. On the other hand **Distributed System** are loosely-coupled system.
- **Parallel computing** is the use of two or more processors (cores, computers) in combination to solve a single problem.

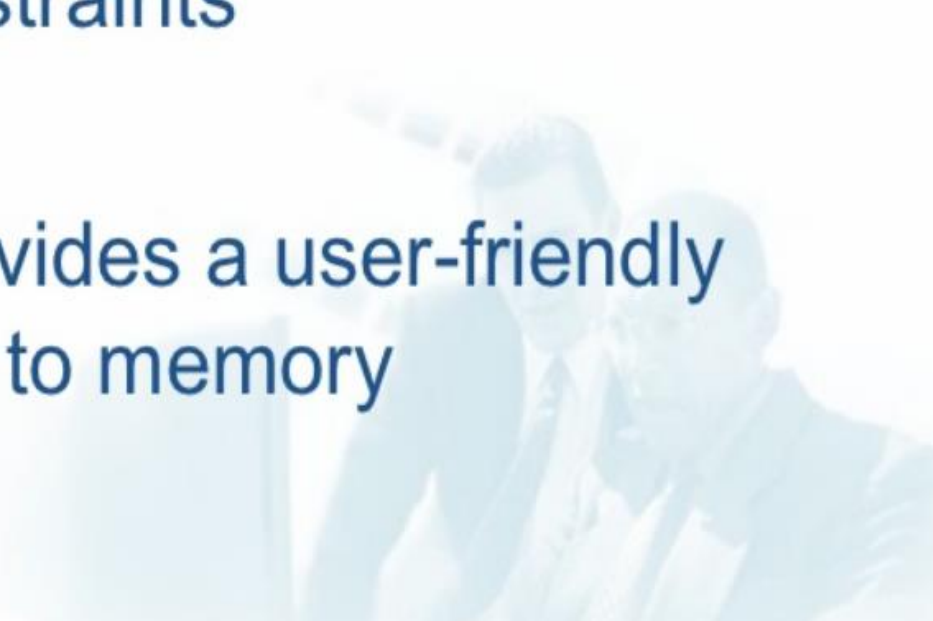


A Parallel System





Advantages of Parallel System

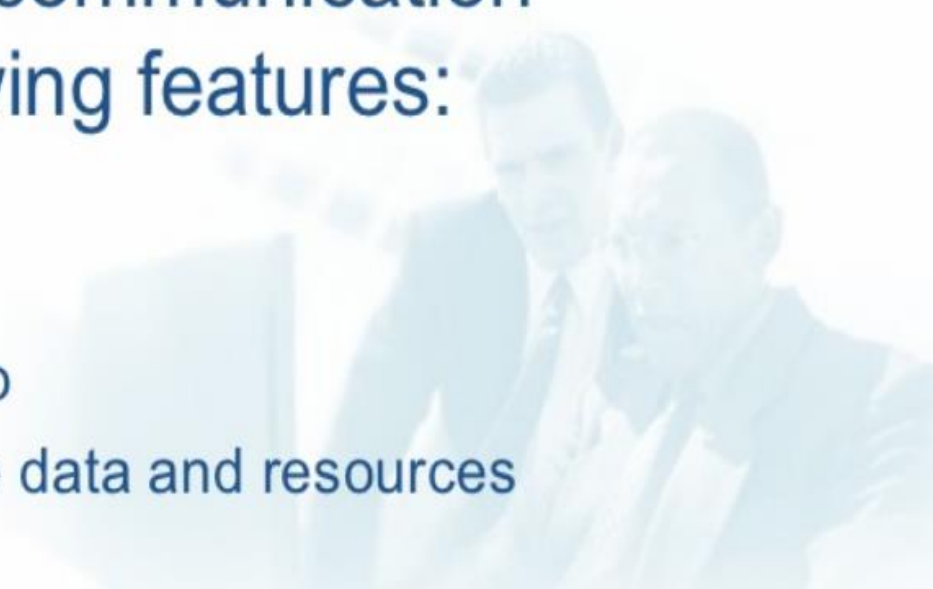
- Provide Concurrency(do multiple things at the same time)
 - Taking advantage of non-local resources
 - Cost Savings
 - Overcoming memory constraints
 - Save time and money
 - Global address space provides a user-friendly programming perspective to memory
- 



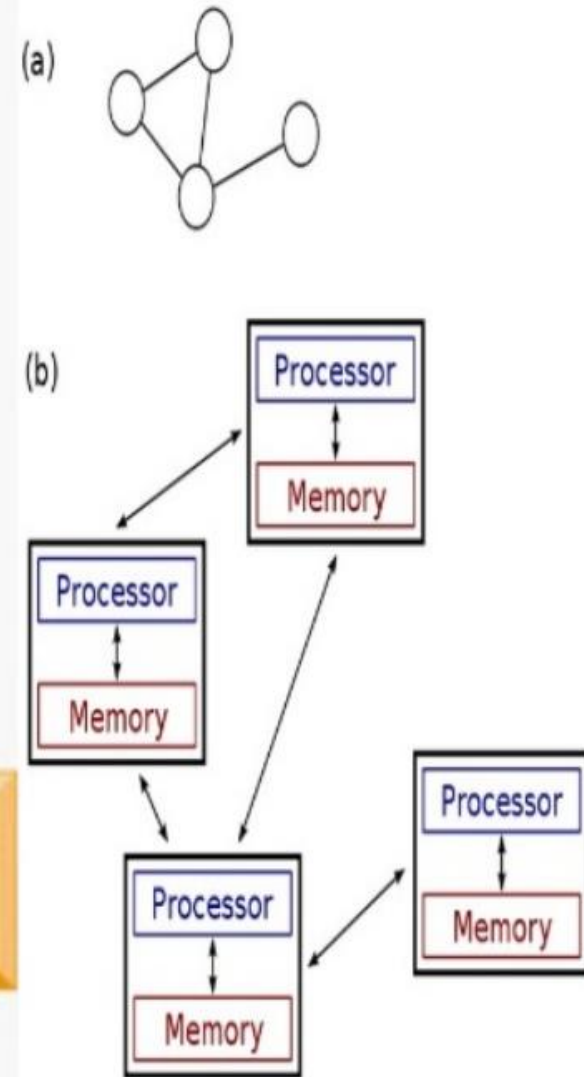
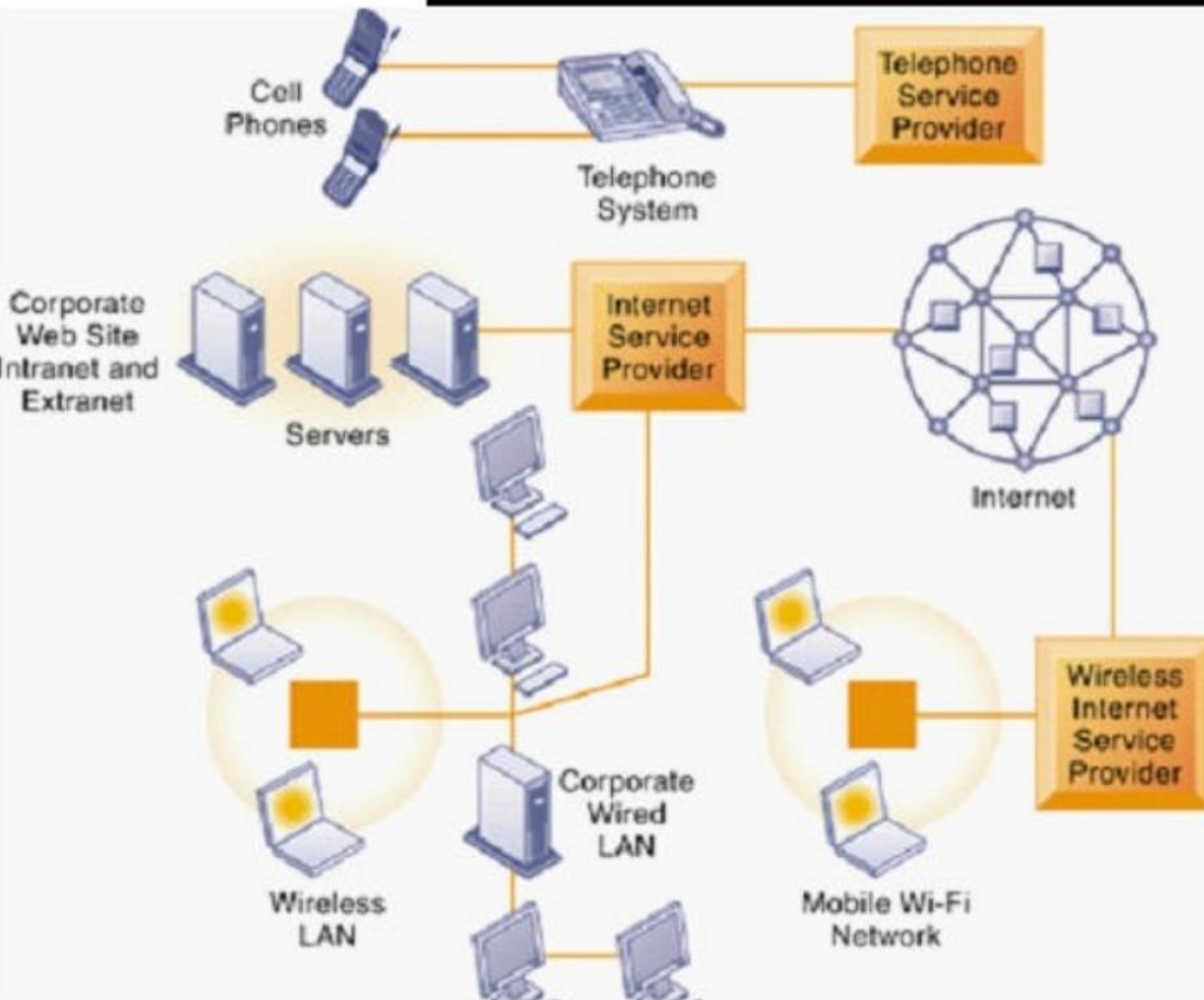
Disadvantages of Parallel System

- Primary disadvantage is the lack of scalability between memory and CPUs.
- Programmer responsibility for synchronization constructs that ensure "correct" access of global memory.
- It becomes increasingly difficult and expensive to design and produce shared memory machines with ever increasing numbers of processors.

Distributed System:

- A **distributed system** is a collection of independent computers, interconnected via a network, capable of collaborating on a task.
 - A **distributed system** can be characterized as collection of multiple autonomous computers that communicate over a communication network and having following features:
 - ✓ No common Physical clock
 - ✓ Enhanced Reliability
 - ✓ Increased performance/cost ratio
 - ✓ Access to geographically remote data and resources
 - ✓ Scalability
- 

A Distributed System





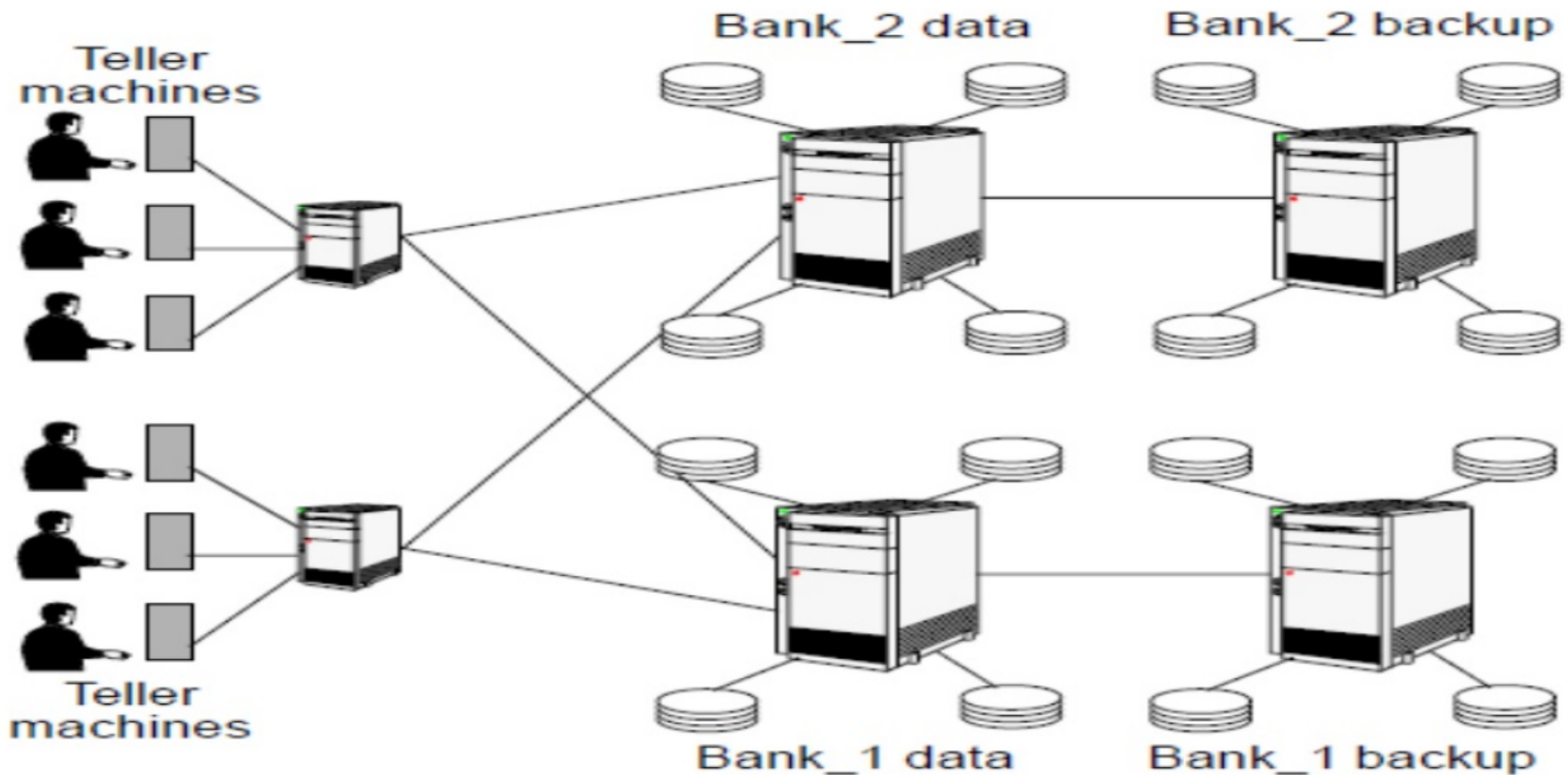
Examples of Distributed System

- Telephone Networks and Cellular Networks
- Computer Networks Such as internet or intranet
- ATM(bank) Machines
- Distributed database and distributed database management system
- Network of Workstations
- Mobile Computing etc.



Examples of Distributed Systems (cont'd)

Automatic banking (teller machine) system



- Primary requirements: security and reliability.
- Consistency of replicated data.
- Concurrent transactions (operations which involve accounts in different banks; simultaneous access from several users, etc).



Advantages Of Distributed System

- Information Sharing among Distributed Users
 - Resource Sharing
 - Extensibility and Incremental growth
 - Shorter Response Time and Higher Output
 - Higher Reliability
 - Better Flexibility's in meeting User's needs
 - Better price/performance ratio
 - Scalability
 - Transparency
- 



Disadvantages of Distributed System

- Difficulties of developing distributed software
- Networking Problem
- Security Problems
- Performance
- Openness
- Reliability and Fault Tolerance



Parallel vs. Distributed System

Parallel Systems

Distributed Systems

Memory

Tightly coupled system
shared memory

Weakly coupled system
Distributed memory

Control

Global clock control

No global clock control

Processor interconnection

Order of Tbps

Order of Gbps

Main focus

Performance
Scientific computing

Performance(cost and scalability)
Reliability/availability
Information/resource sharing



6. Shared Nothing architecture

➤ Three most common types of architecture for multiprocessor high transaction rate systems are:

➤ **a. Shared Memory (SM)**

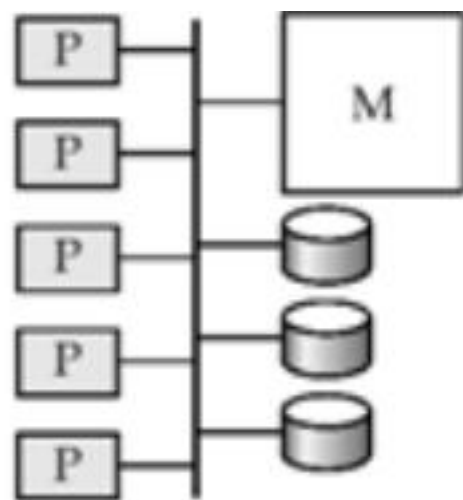
--- A common central memory is shared by multiple processors.

➤ **b. Shared Disk (SD)**

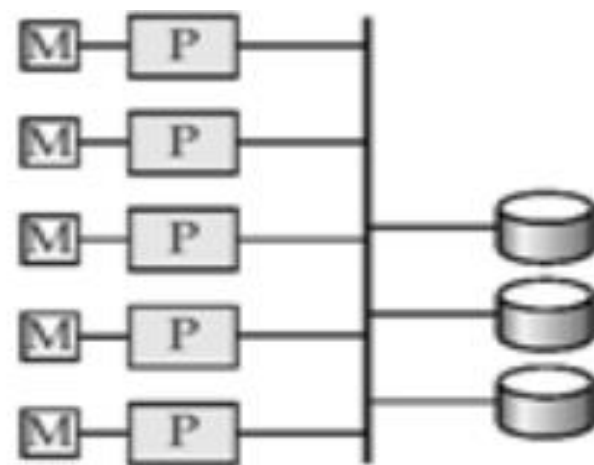
--- Multiple processors have a common collection of disks having their own private memory.

➤ **c. Shared Nothing (SN)**

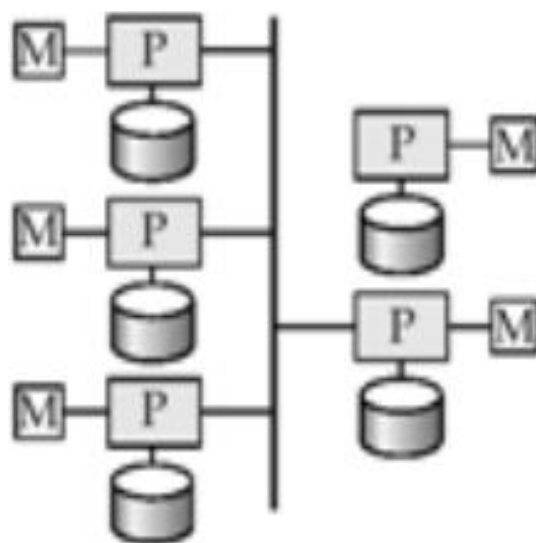
--- Neither memory, nor disk is shared among the multiple processors.



(a) shared memory



(b) shared disk



(c) shared nothing

Advantages and Disadvantages of SN Architecture

Advantages

- › **Elimination of SPOF (single point of failure)** – if a part of a system fails, it won't stop the entire system from working
- › **Non-disruptive upgrades** – database updates will not interrupt access to database data
- › **No need for reboots** – SN does not require system as a whole to be rebooted when the update process completes.
- › **Self-healing capabilities** – allows a database to be repaired while still processing transactions

Disadvantages

- › Hard to program
- › Addition of new nodes requires reorganizing

7. CAP Theorem

It is also called as *Brewer's Theorem* or *two out of three theorem*.

“It states that in a distributed computing environment, it is impossible to provide the following guarantees. At the best you can have two of the following three – one must be sacrificed.”

1. Consistency

2. Availability

3. Partition Tolerance

Example: Updating a status in a Facebook.

We do not want any type of consistency and also we do not want that our whole servers are available because it may be happens that it is available to some people.

Consistency – This means that the data in the database remains consistent after the execution of an operation. For example after an update operation all clients see the same data.

Availability – This means that the system is always on (service guarantee availability), no downtime. **Partition Tolerance** – This means that the system continues to function even the communication among the servers is unreliable, i.e. the servers may be partitioned into multiple groups that cannot communicate with one another.

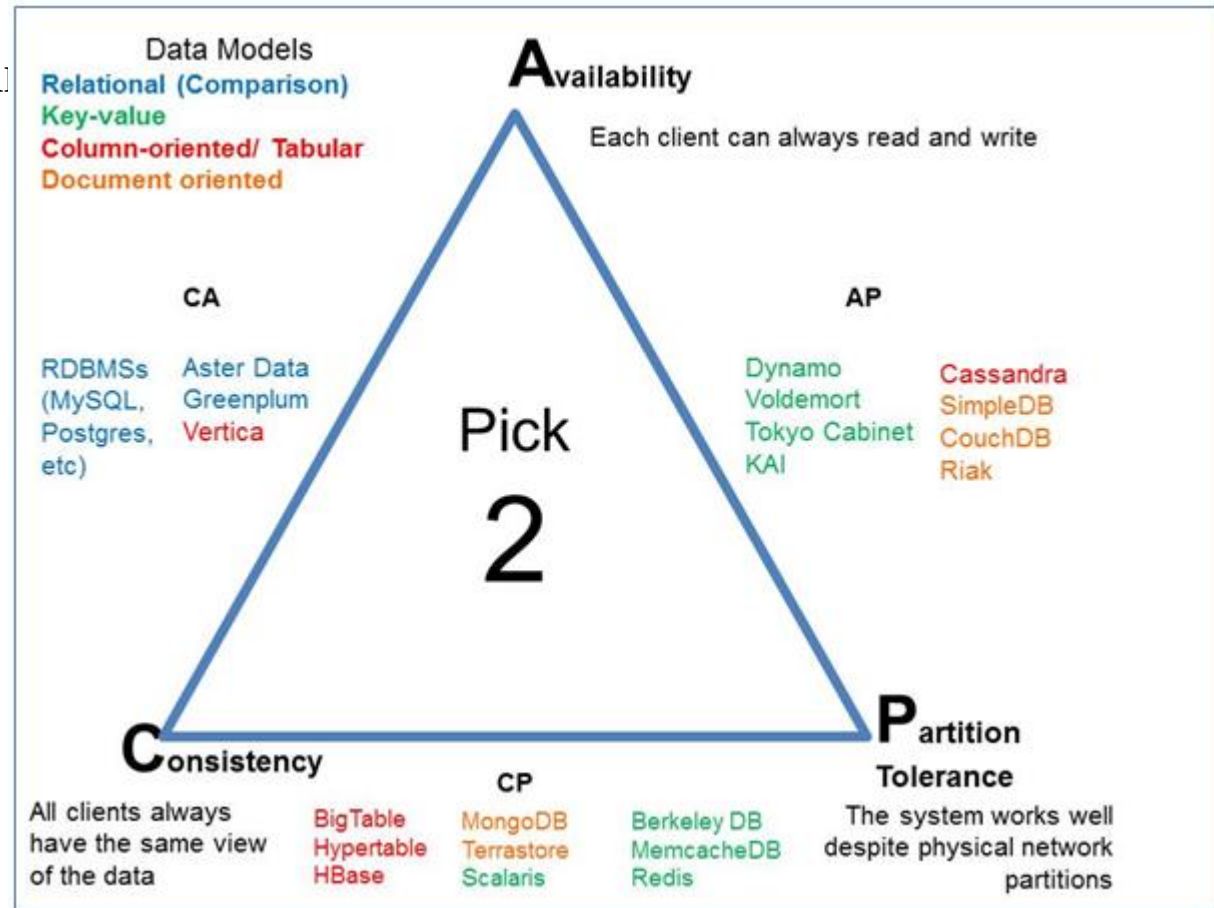
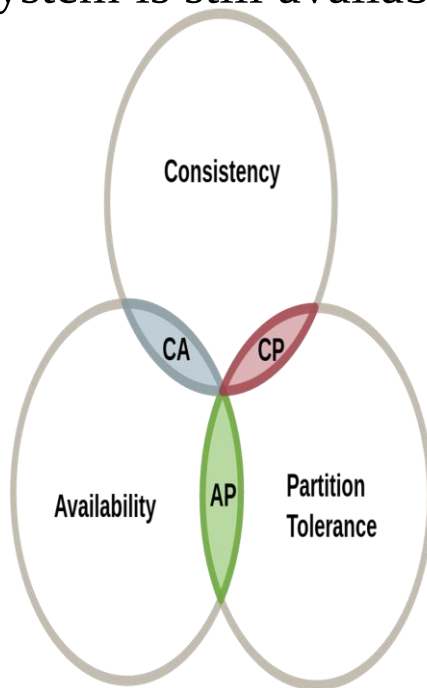
- › Theoretically it is impossible to fulfill all 3 requirements.
- › CAP provides the basic requirements for a distributed system to follow 2 of the 3 requirements.
- › Therefore all the current NoSQL database follow the different combinations of the C, A, P from
- › the CAP theorem.

Here is the brief description of three combinations CA, CP, AP :

CA – Single site cluster, therefore all nodes are always in contact. When a partition occurs, the system blocks.

CP – Some data may not be accessible, but the rest is still consistent/accurate.

AP – System is still available under partition



Summary of CAP Theorem with Case study

One can atmost go with two of the three.

1. Consistency:

→The instructors or training coordinator, once they have updated information with you, will always get the updated information when they call subsequently.

2. Availability

→The instructors or the training coordinators will always get the schedule if any or both of the office administrators have reported to work.

3. Partition Tolerance

→Work will go on as usual even if there is communication loss between the office administrators

→owing to a spat (past) or a tiff(quarrel)

BASE PROPERTIES

- BASE stands for Basically Available, Soft state and Eventually consistent.
- Based upon CAP theorem, NoSQL has BASE properties.
- Based upon that BASE properties it works in a manner that it may be eventually consistent having soft state when it is basically available.

➤ **Basically Available:**

- When data is stored at one site, If that site is down, our data is totally unavailable but if our data is distributed to many sites, means some part of our data is at one site and some part of data is at other site.
- If one or two sites down our whole data is not unavailable.
- It may be that some part of data becomes unavailable for some short time

Soft State:

Consistency is hard requirement of ACID properties but BASE allows that our data may be inconsistent.

It allows dynamic designing to application developers.

Eventually Consistent:

NoSQL systems assume that they provided consistency in the future, not immediately at the end of operation.

In contrast to ACID systems that enforce consistency after transaction commit,

NoSQL guarantees consistency only at some undefined future time.

So eventually consistency means it is consistent after some time even at that time it is not guarantee of the consistency.

ACID vs BASE PROPERTIES

ACID	BASE
It stands for Atomicity, Consistency, Isolation and Durability	It stands for Basically Available, Soft state and Eventually consistent
These are pessimistic that means it guarantees that consistency should be achieved at the end of any operation.	These are optimistic means that it provides no guarantees that consistency should be achieved at the end of any operation.
Basic requirement in RDBMS	Basic requirement in NoSQL
These are complex to implement all properties	These are simple to implement all properties
Provides reliability	Provides no reliability

When to choose consistency over availability and vice-versa....

1. Choose Availability over consistency when your business requirements allow some flexibility around when the data in the system synchronizes.
2. Choose Consistency over Availability when your business requirements demand atomic reads and writes.